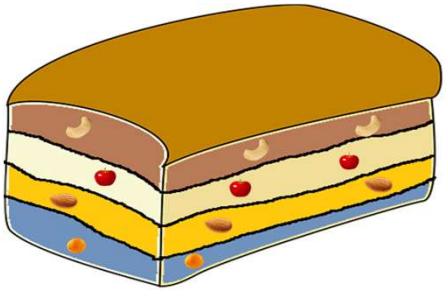
Network Slicing in 5G

What all are covered

- Need for Network Slicing in 5G
- What is Network Slicing in 5G
- Network Slicing – Enablers in 5G
- Realizing Slices in 5G Network Subsystems
- Implementation of End-to-End Slices in 5G
- Network Slice Management
- Few Use cases

What is a Network Slice?



What is a Network Slice?

- Network slicing is a form of virtualization that allows multiple logical networks with specific network capabilities and network characteristics to run on top of shared physical infrastructure.
- Network Slicing allows a physical mobile network operator to partition the network resources to allow for very different users (called tenants) and services, to multiplex over a single physical infrastructure.

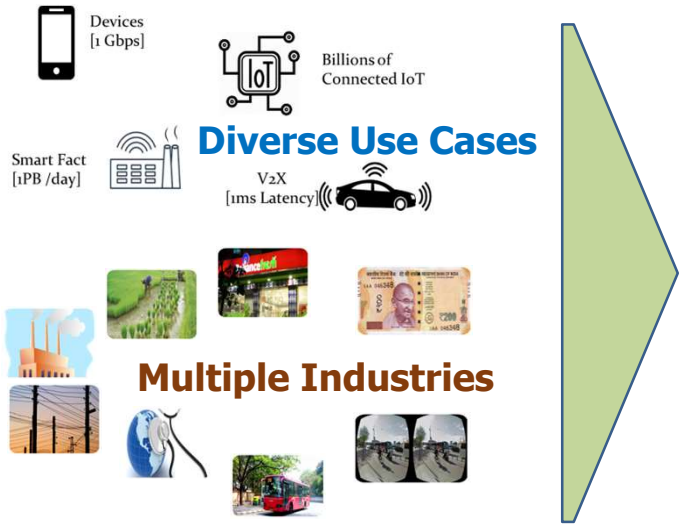
Need for Slicing in 5G

Variable Service requirements

	Throughput	Coverage	Latency	Number of Devices
Water meter	Very low	Large	High	Large
Assisted driving	Low	Medium	Very Low	Low
Smart phone users	Medium	Medium	Medium	Medium
Virtual reality (Online Gaming)	Very High	Medium/Low	Low	Low
Fixed wireless	High	Medium to Large	Medium	Medium
Remote diagnostics	Medium	Medium	Medium to High	Medium
Remote surgery	High	Low	Very low	Low

How to meet these diverse requirements from a Single network?

Demands from 5G Networks



Diverse Performance

Diverse Service Requirements

Diverse Customers

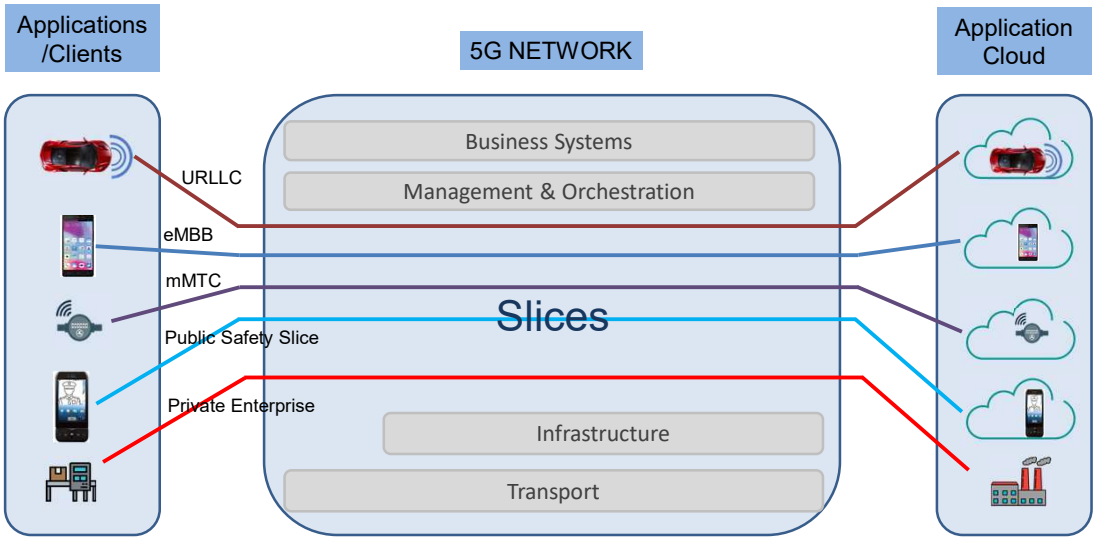
Scalability, Security, Service Agility, Cost

Needs from Slicing to provide Customized Networks

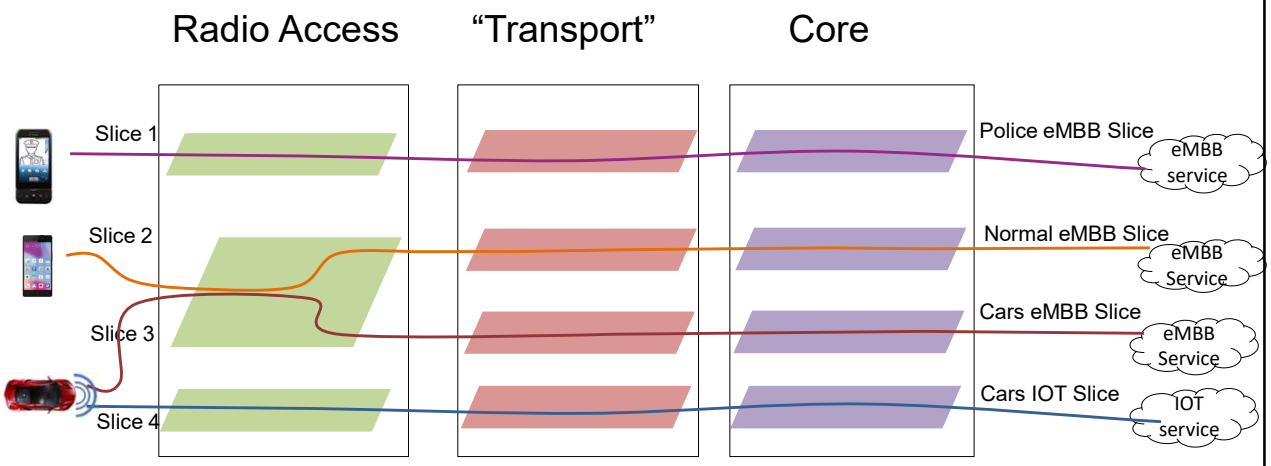
- Different Functionality Requirements
  - Priority, Policy, Mobility, Security etc
- Different Performance Requirements
  - Latency, Data rates, Mobility, Reliability etc
- Different Users
  - Public Safety users, Enterprise customers, MVNO etc

What is Network Slicing in 5G

Slices in 5G Network

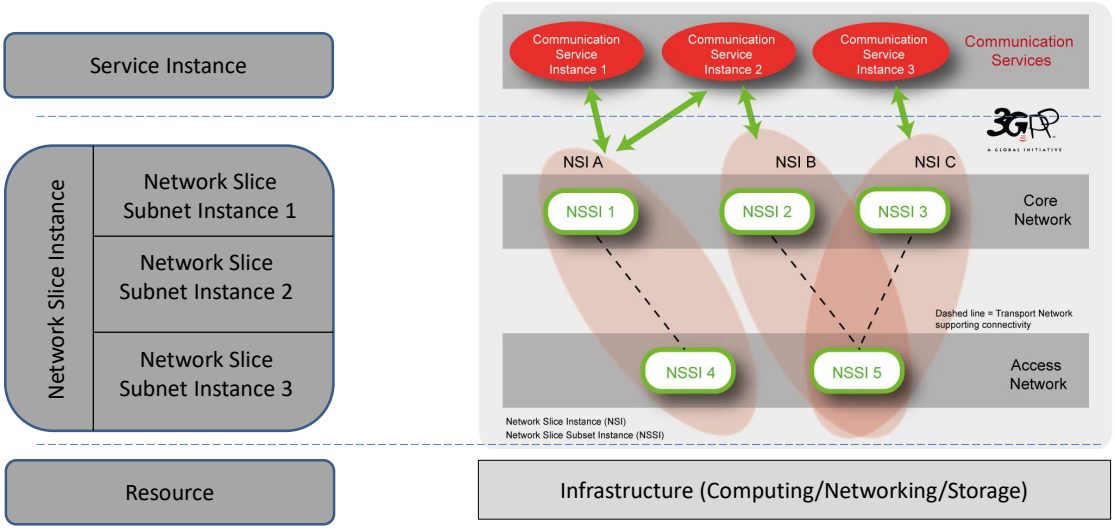


End to end Slice through RAN, Core and Transport



- A UE can access multiple slices in the Network
- A Slice can service multiple UEs and Services
- A Slice is realised at RAN, Core and Transport

Layered perspective of Network Slices

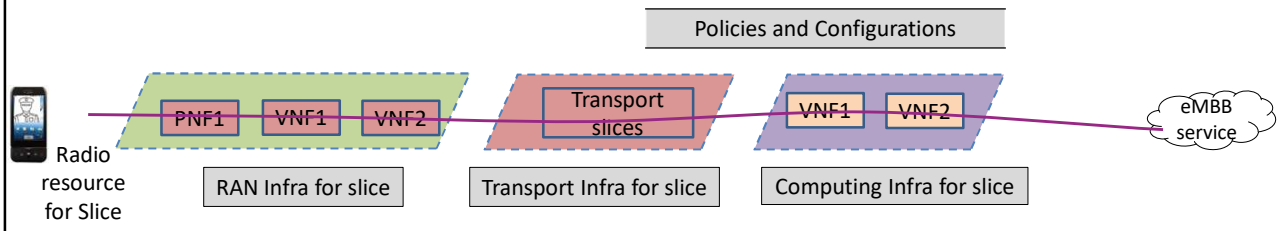


SI, NSI, NSSI

- Service Instance :
  - The end-user or business services, provided by a network operator, to be supported by the slice. Represented by a Service Instance (SI).
- Network Slice Instance :
  - A set of NF instances and the required resources (e.g. compute, storage and networking resources) which form a deployed Network Slice.
  - Network Slice can be deployed in a part of a network.
  - A Network Slice Instance can be shared among multiple Service Instances.
- Network Slice Subnet Instance :
  - Network Slice Instance may include one or more Sub-Network Instances to form a set of Network Functions running in physical or logical resources.

What is a Network Slice composition?

- A set of network functions
- Resources to run these network functions
- Policies and configurations
- Specific configurations of transport network
- Dedicated radio configuration or even a specific RAT
- Configuration of the 5G device



Key Features of Network Slicing in 5G

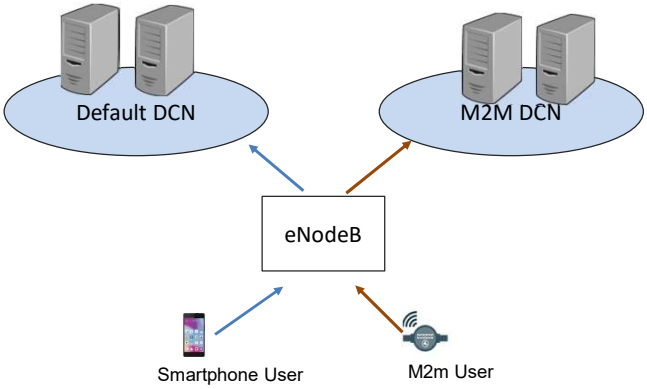
- Slicing can be end-to-end or part of network
  - User equipment, Access network, Core network, Transport network, Cloud, Application servers
- Dedicated and / or shared resources at every sub-system
  - Tradeoff
- Flexible, customizable, easily built and teared down
- Elastic and Extensible - Scaling Up/Down on-demand, Adding functionality
- Slices as managed network with required QoS
- Slices can be isolated from each other -> increasing safety and security of slices
- Slice exposure – additional authentication

Benefits brought by Slicing

- Widens the Customer Base for the Operators / Infrastructure Providers
  - Slices for Vertical Applications
  - Slices for diverse Horizontal services
  - Non-public networks
- New business models
  - Ex. Virtual Networks for virtual operators
- Optimising of network resources
  - Right Sizing
- Network as a service
  - On demand slices!
- And More as the concept evolves……..

What was there in 4G?

- Called DECOR/eDECOR (R13, R14) - Dedicated Core Network (DCN)
- Multiple Dedicated Core for networks within a PLMN
- Each DCN consisting of one or multiple CN nodes
- Each DCN may be dedicated to serve specific types of subscribers



Network Slicing - Enablers in 5G

Enablers for Slicing

Network Function Virtualisation (NFV)

Software Defined Networks (SDN)

Software Defined Radios

Cloudification

NFV enabling Slicing

What is NFV?

- Decouple software (function) and hardware
- Hardware shared across Network Functions
- NFs implemented as software and run on VMs/ as containers
- Power of slicing and orchestration using the flexibility brought in by VNFs

NFV in Slicing

- Provides the flexibility for sharing resources among Network Slices
- VNFs may be shared across slices
- Enables deploying and scaling Network Slices automatically.

SDN enabling Slicing

What is SDN?

- Separation of Control plane and Data plane via standard interfaces
- Allows certain NFs to handle the common control plane functions while having dedicated NFs to handle data plane related functions

SDN in Slicing

- Segregating Common control functions and Session related functions
  - Ex. Common RRC, but different SDAP, PDCP and RLC
- Slice hierarchy managed by corresponding control plane hierarchy
  - Ex. Network Sub network Slice in RAN by RAN Controller
- A tool for efficient and automatic management of Slices

SDR enabling RAN Slicing

What is SDR?

- Uses software for Radio Signal processing instead of traditional way of using hardware
- General purpose device or reconfigurable digital piece electronics used for many waveforms

SDR for Slicing

- Flexibility of allocating Frequency bands, Bandwidth parts, Carrier Aggregation etc
- Flexibility in allocation of Radio Resources, Coding, Modulation
- Flexibility in configuring Antenna, Beam forming etc

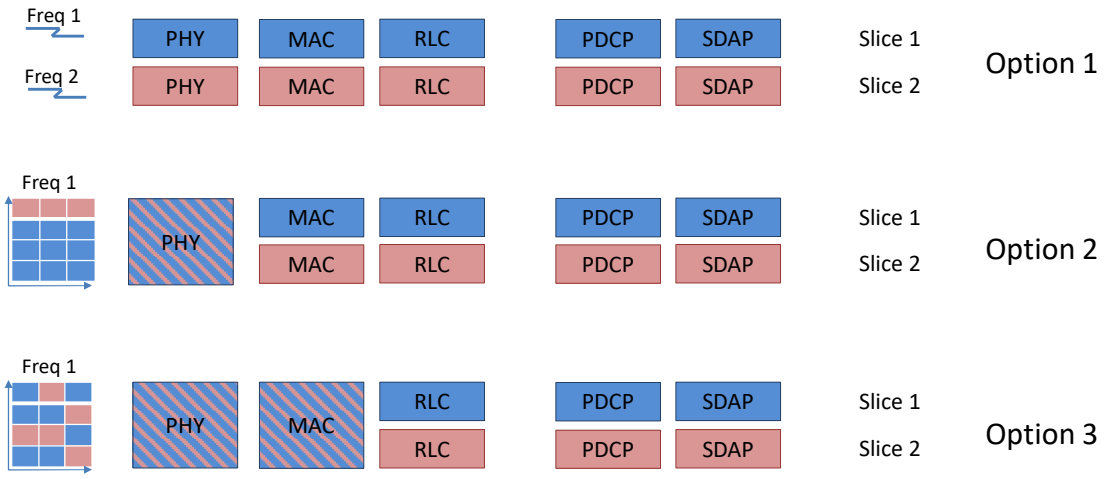
Realizing Slices in 5G Network Subsystems

Typical Subsystems sliced as NSSIs

- RAN Slicing
  - AN stacks – Control plane & User Plane
  - Radio resources
  - MAC schedulers
- Transport and Switching Layer Slicing
  - All intermediate components within the link
  - QoS configurability
- Core Network Slicing
  - Multiple instances and/or shared NFs

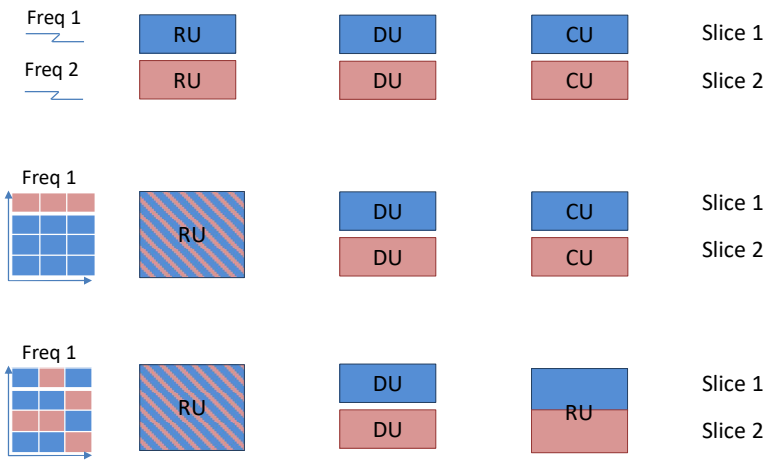
RAN Slicing

Examples of providing Slicing in RAN



RAN Slicing

Examples of Slices handled by various RAN Nodes



RAN Slicing

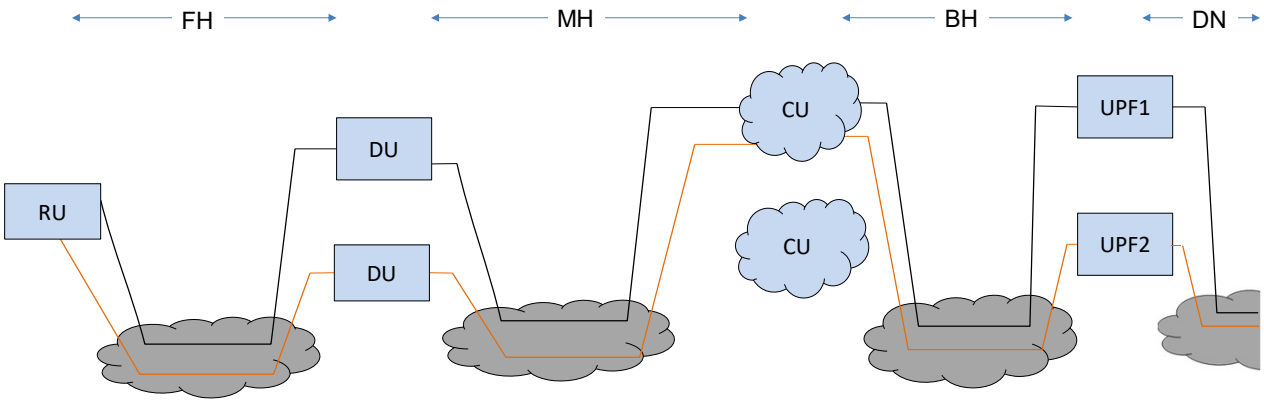
- RAN can realize slices by scheduling and by providing different L1/L2 configurations.
- How NG-RAN supports the slice enabling in terms of NG-RAN functions is implementation dependent.
- Supports resource isolation between slices. How it does is implementation dependent.
- NG-RAN is configured with a set of different configurations for different network slices by OAM
- RAN Slicing is an advanced form of RAN Sharing
  - Radio resource scheduling can be with the Virtual operator. The RAN Infra provider can be transparent to the business aspects of the radio scheduling.
- Slice may be accepted/rejected based on support for the slice, availability of resources, support of the requested service by NG-RAN

Transport Slicing

- Work being done in IETF on Transport Slicing

  IETF Definition: A Transport Slice is an abstract network topology connecting different endpoints with appropriate isolation and specific Service Level Agreement (SLA) described in terms of shared or dedicated network resources, level of isolation etc.
- Abstract definition and realization of Transport Slices are separated
  - Orchestrator not aware of the transport Technologies, Tunnel types etc
- Transport Slice Controller
  - Provides an abstract, programming interface to End to End Slice Orchestrator to create and manage transport slices
- Hard slicing (dedicated resources) and Soft Slicing

Transport Slicing

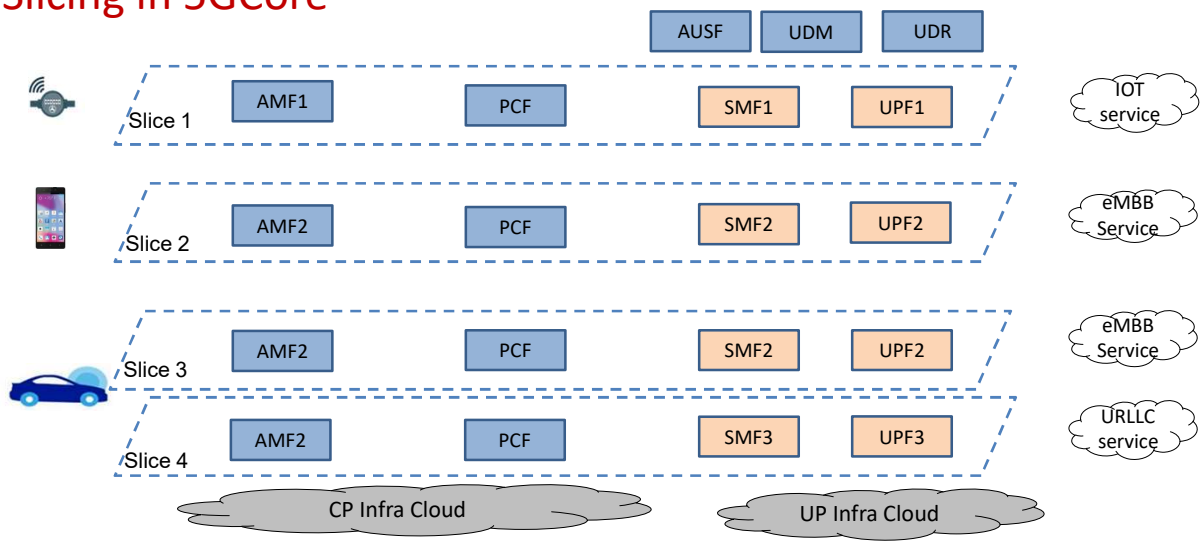


- Transport slices across Fronthaul, Midhaul, Backhaul and DN connectivity

Transport Slicing

- Improves Service Layer Adherence
- Simplifies operation of multi-domain, multi-technology, multi-vendor transport networks
  - Breaks to manageable pieces
- Segment Routing,L3VPN (MPLS VPN) etc can enable slicing in transport networks

Slicing in 5GCore



- Each of the 5GCore NF (CP and UP) can be associated with one or more slices
- Different PDU Session NFs are generally allocated for Slices

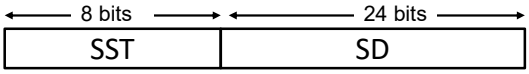
Implementation of End to end Slices in 5G Networks

Principles of Slicing

- Each network slice in a PLMN is uniquely identified by a value called S-NSSAI
- Network can realize the different network slices by configuring various NFs that form part of the Slice.
- Traffic for different slices is handled by different PDU sessions - Granularity of slice awareness at PDU Session level
- Supports QoS differentiation within a slice – 5G QoS Flows
  - Same 5QI flows across Slices can be treated differently
- Network Slices supported by each NF is configured by OAM.
- Apps in UEs are associated to specific slices through NSSP (Network Slice Selection Policy)

Slice Identifier

- Uniquely identifies an End to end Network Slice across UE, RAN and Core
- Slice identified by Single Network Slice Selection Assistance Information (S-NSSAI)



  - SST (Slice/Service Type) –> (Indicative) Network Slice behavior. Standard values (Rel16) are:

Slice/Service type	SST value	Characteristics
eMBB	1	Slice suitable for the handling of 5G enhanced Mobile Broadband.
URLLC	2	Slice suitable for the handling of ultra- reliable low latency communications.
MIoT	3	Slice suitable for the handling of massive IoT.
V2X	4	Slice suitable for the handling of V2X services.

  - SD - Slice Differentiator (Optional) - Differentiates among Slices with same SST
  - Standardized S-NSSAI has only SST and no SD
  - Non-Standard S-NSSAI can be defined as either SST alone or SST + SD
- NSSAI – set of S-NSSAIs send slice info in the signalling messages
- Maximum 8 S-NSSAIs in NSSAI -> max 8 slices accessible by a UE
- S-NSSAI and NSI ID are configured by the management system

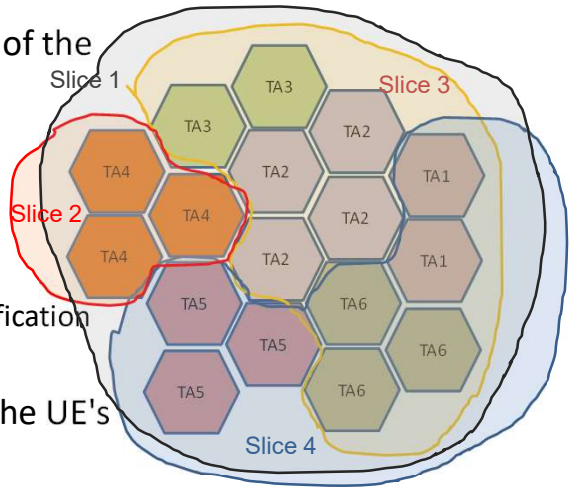
Different forms of NSSAIs

- Configured NSSAI: NSSAI provisioned in the UE applicable to one or more PLMNs
  - Default Configured NSSAI configured by the HPLMN
- Subscribed S-NSSAIs: S-NSSAIs based on subscriber information, which a UE is subscribed to use in a PLMN.
  - One or more can be default S-NSSAIs
- Requested NSSAI: NSSAI provided by the UE to the Serving PLMN during registration
- Allowed NSSAIs: S-NSSAIs that UE is allowed to access in the Registration Area
- Rejected S-NSSAIs: S-NSSAI that are rejected by the Network for a request from UE
- Pending NSSAI: NSSAI provided by network, indicating the S-NSSAI(s) for which the network slice-specific authentication and authorization procedure is pending.

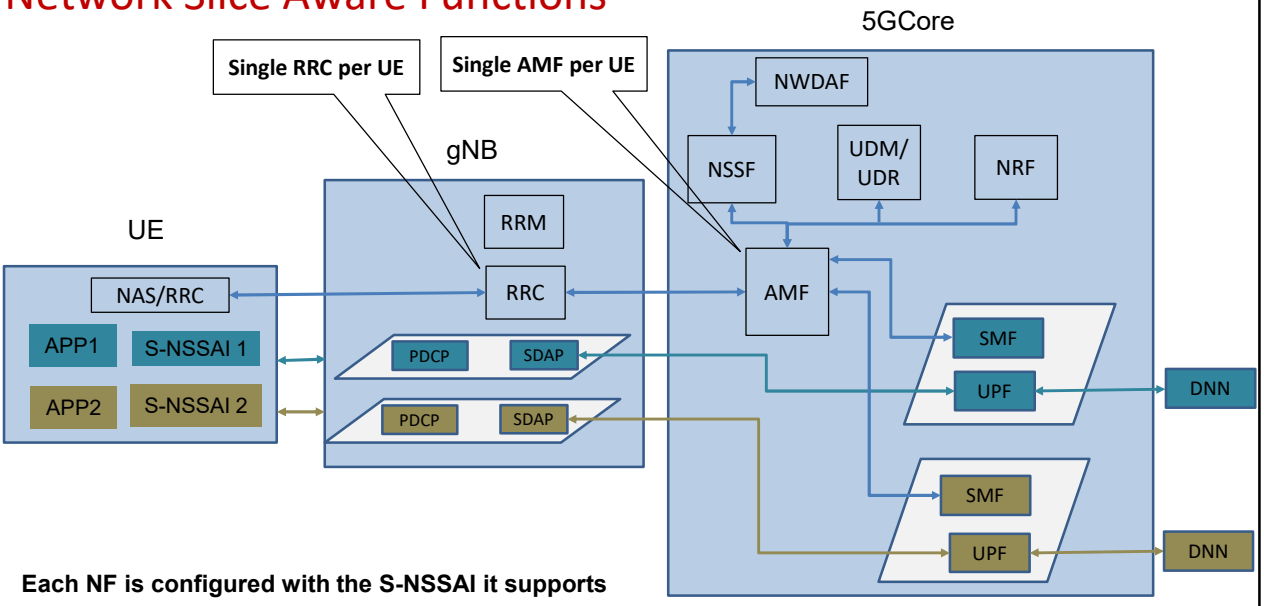
Serving PLMN
Home PLMN

Validity of Slices

- Some slices may be available only in part of the network (TAs)
  - Can be whole PLMN or a part of it.
- PLMNs supporting the slice
  - Part of Roaming Agreements
  - Slices are allowed in Visiting PLMN after verification during Registration
- Slice availability does not change within the UE's registration area



Network Slice Aware Functions



Each NF is configured with the S-NSSAI it supports

Slicing aspects in key procedures

Registration Procedure:

- UE requests and gets approved list of Slices (S-NSSAIs)
- Updated during every registration

PDU Session Procedure:

- UE requests for PDU Session for a specific slice and DNN

NG Setup Procedure:

- gNodeB and 5GCore exchanges Slicing info that each supports

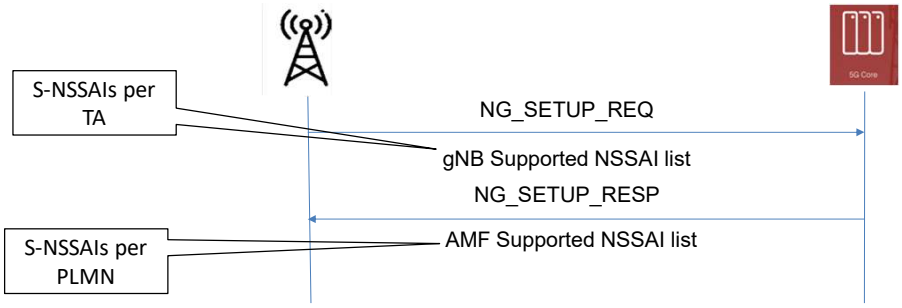
UE Configuration Update Procedure:

- 5GCore provides Slicing info to UE

Slicing info used in other procedures like Service Request also (NAS as well as RRC messages).

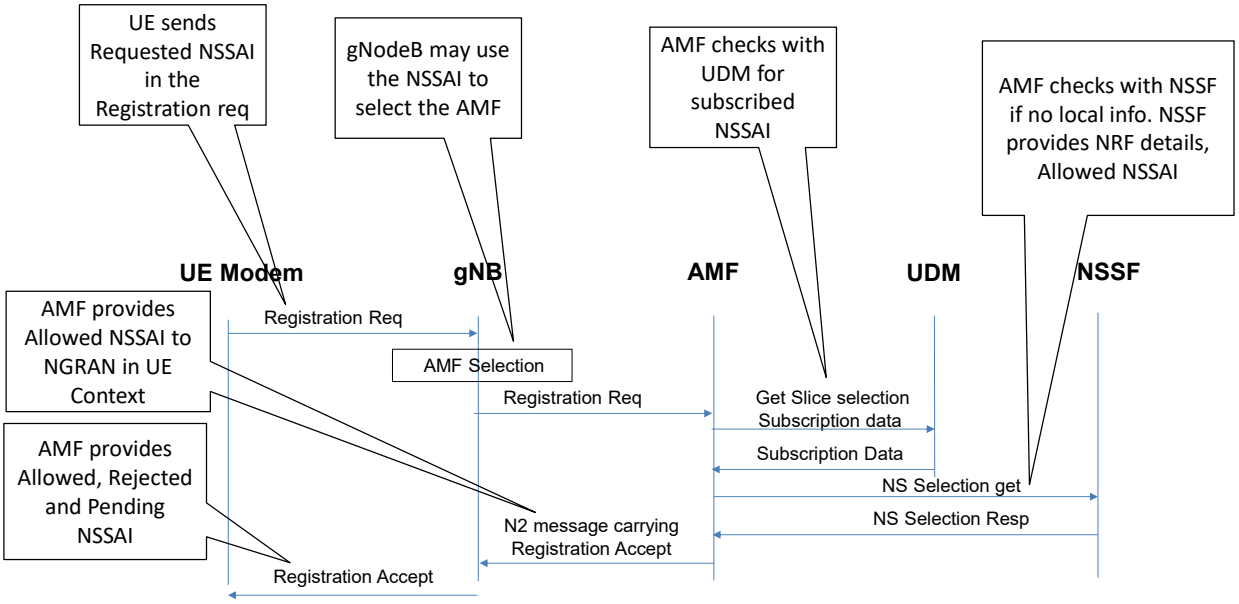
Slicing information exchange in Procedures

- Exchange between AN and Core during NG-SETUP procedures – Slices supported
  - Slice identity important parameter for AMF selection for a UE at NG-AP

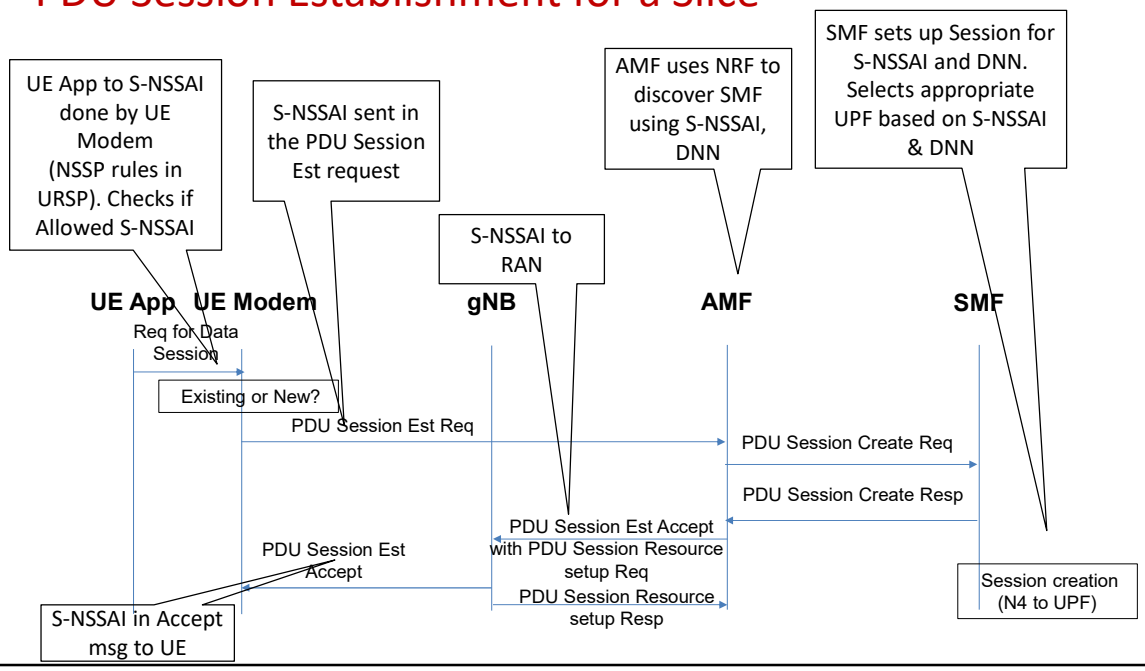


- Exchange between Core and UE
  - UE Configuration Update procedure is used to update NSSAI info in UE - Subscription info changes, Authorization changes etc

Slicing info during UE Registration



PDU Session Establishment for a Slice

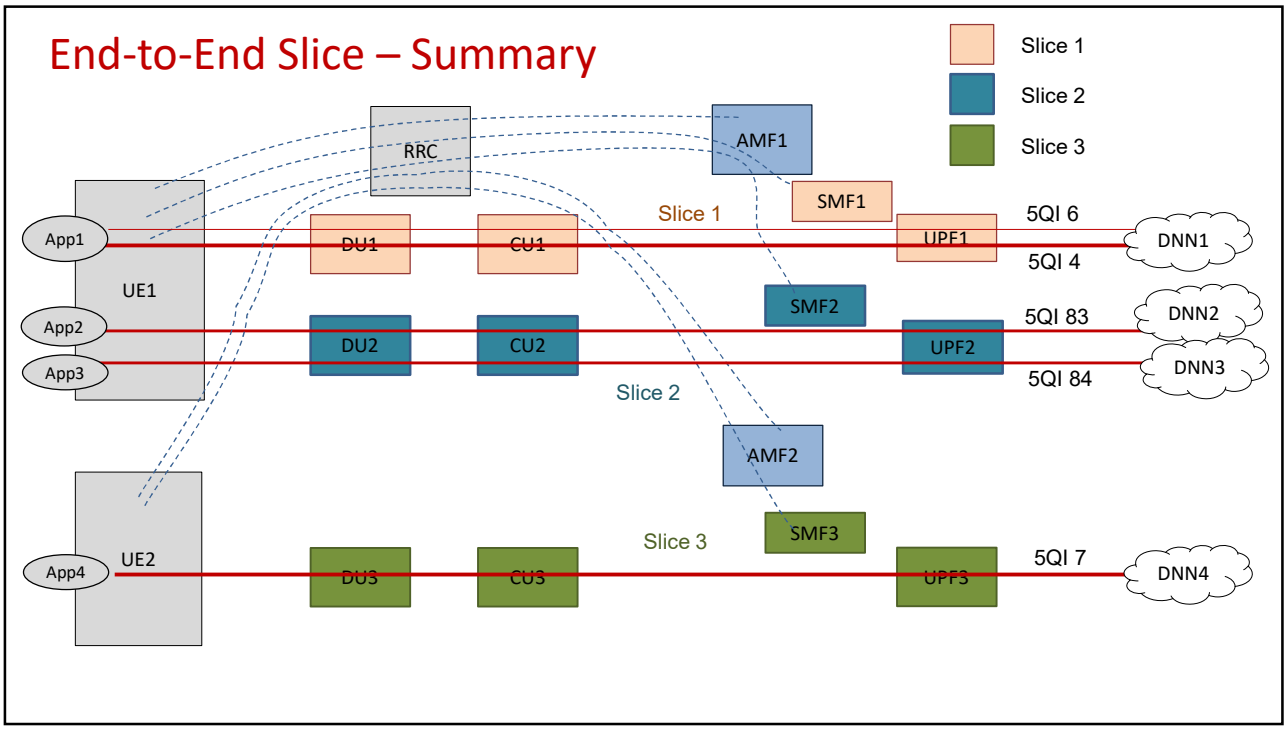


Network Slice Selection Function -NSSF

- AMF may optionally use services of NSSF
- NS Selection Service (Network Slice Selection)
  - Network Slice selection in both the Serving PLMN and the HPLMN
  - Provides Allowed NSSAI, Configured NSSAI, target AMF Set /AMFs, NRF for the slice, NSI Id [O] etc
- NSSAI Availability Service
  - AMF uses the service to update the S-NSSAI(s) supported on a per TA basis
  - NFs can subscribe and get notified of changes to the NSSAI availability information on a per TA basis

Network Data Analytics Function - NWDAF

- Provides slice load level information to an NF (NSSF, PCF) on a network slice instance level.
- Events based on slice load crossing specified threshold
- Used by NSSF for Slice Selection



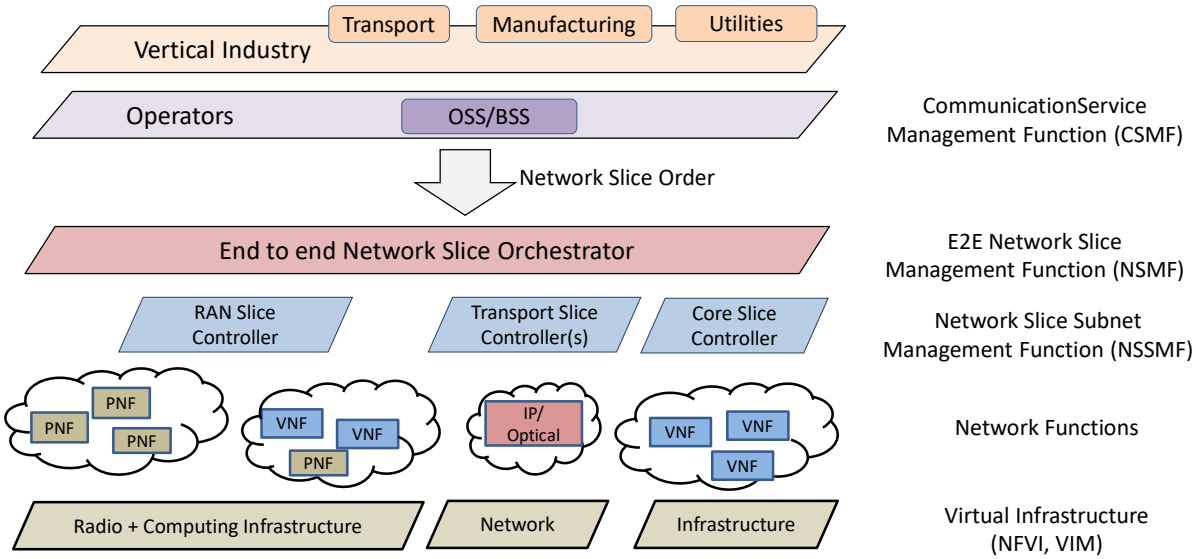
Network Slice Management

Criticality of Network Slicing management

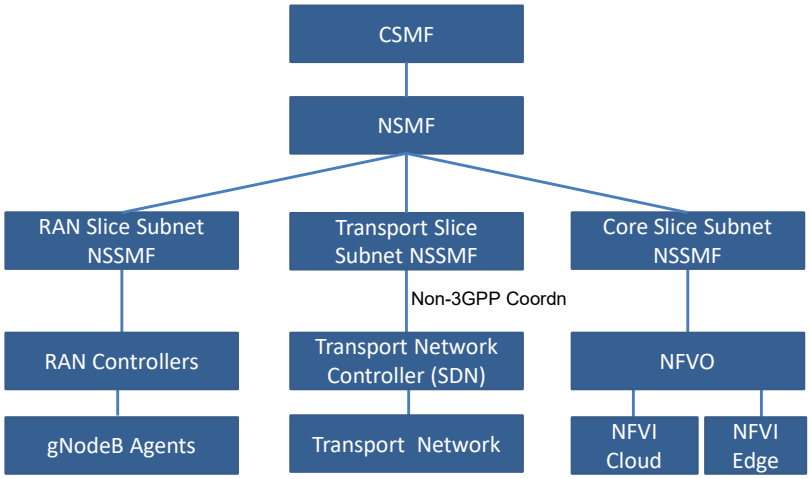
- Multiple slices created to provide various eMBB/URLLC/IOT Services over the 5G Network
  - Running multiple virtual network is a management challenge
- Network Slicing has management overheads =>

  Automation of Slice creation and management
- Operators can track and scale the capacity of each slice
  - Optimizes network utilization. Start small and grow as the demand grows
- Network Slices can be created dynamically – new in mobile networks

Network Slicing management Architecture



Slice Management Framework



Communication Service Management Function (CSMF):
Translate the communication service

Network Slice Management Function (NSMF):
For management of NSI.

Network Slice Subnet Management Function (NSSMF):
Domain slice managers.

Domain/SDN Controllers

Infrastruture / Agents

Creating Network Slice

Service Template

1. Slice Type: URLLC
2. Slice performance
   - Throughput: 10 Gbps
   - Latency : 5ms
   - Reliability : 99.99%
3. Capacity
   - No of UEs: 1000
4. Functionality :
   - Broadcast

Customer

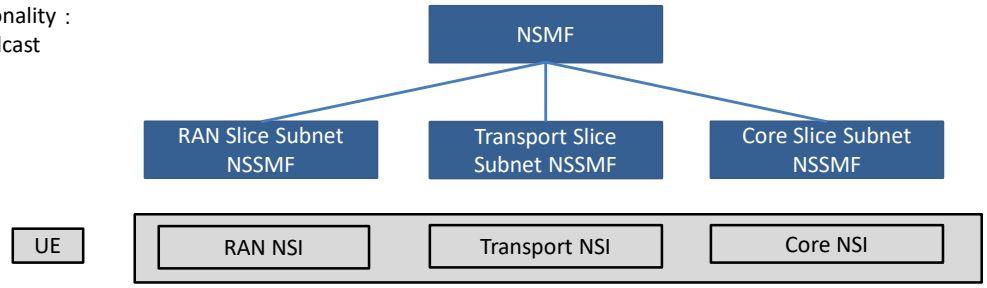
Service Requirements using Service Templates

GST (Generic Slice Template)

Operator

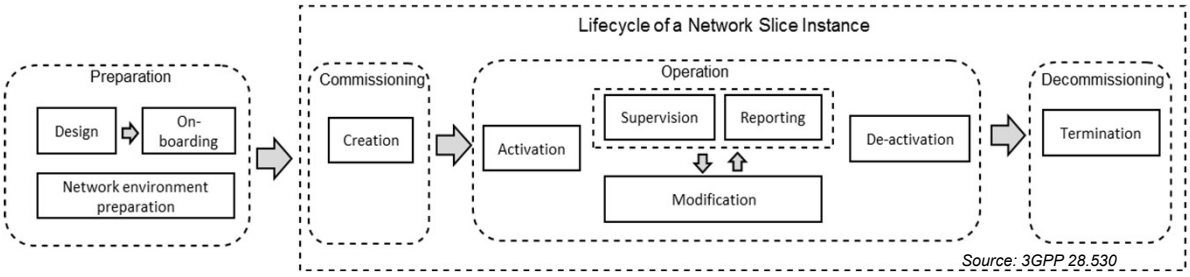
Service Template to Network Slice template NEST (Network Slice Type) created.

Slice Instantiation



S-NSSAI = RAN NSSI + Transport NSSI + Core NSSI

Network Slice – Life Cycle Management (NSI/NSSI)

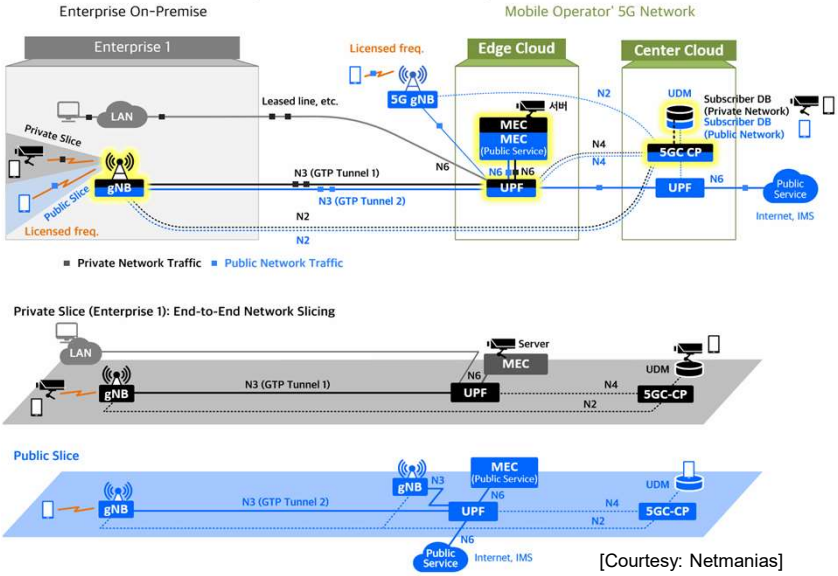


Preparation → **Commissioning** → **Operation** → **Decommissioning**

- **Preparation phase** - Creation & Verification of network slice template(s), **NSI does not exist** .
- **Commissioning phase** - All resources shared/dedicated to the NSI created, configured. **NSI ready.**
- **Operation phase** - Activation, supervision, performance reporting (e.g. for KPI monitoring), resource capacity planning, modification, and de-activation of an NSI.
- **Decommissioning phase** - Deactivation and reclamation of dedicated resources (e.g. termination or re-use of network functions) and re-configuration of shared/dependent resources. **NSI no more exists**

THANK YOU

RAN and Core Sharing (End-to-End Network Slicing) between private and public network



[Courtesy: Netmanias]

- Shared CP and UP by Private and Public Slices
- Isolation of Slices can be provided as required